

Q1)

Ans) Here we are penalizing FN instances harsher than FP instances. So out of 3 given metrics (Accuracy, Precision and Recall) we must choose the one which lay emphasis on FN — (0.5)

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN}$$

$$\text{Precision} = \frac{TP}{TP+FP}$$

$$\text{Recall} = \frac{TP}{TP+FN} \quad \text{--- (0.5)}$$

we clearly see it's the recall metrics which lay emphasis on FN (less the FN implies higher Recall). Thus, choose recall.

Q2)

Ans)

Model \ Metric	f	f _{new}
Precision	P	2P
Recall	R	R/2

$$(\text{F1 score})_f = \frac{2}{\frac{1}{P} + \frac{1}{R}} \quad \text{and similarly } (\text{F1 score})_{f_{\text{new}}} = \frac{2}{\frac{1}{2P} + \frac{1}{R/2}} \quad \text{--- (0.25)}$$

or

$$\Rightarrow (\text{F1 score})_f = \frac{2PR}{P+R} \quad \text{and } (\text{F1 score})_{f_{\text{new}}} = \frac{2}{\frac{R+4P}{2PR}} = \frac{4PR}{R+4P} \quad \text{--- (0.15)}$$

or

We can rewrite $(\text{F1 score})_{f_{\text{new}}}$ as $\frac{2PR}{\frac{R}{2} + 2P}$

Since $\frac{R}{2} + 2P \neq P+R$ in every scenario, thus $(\text{F1 score})_f \neq (\text{F1 score})_{f_{\text{new}}}$ — (0.5)

Q3)

Ans) Given :-

- 1) e is residual
- 2) $f(x;w) = w^T x$ (predicted value)
- 3) y is actual / target value
- 4) $e = y - f(x;w)$
- 5) $e \sim N(0, \sigma^2)$ (Initially)

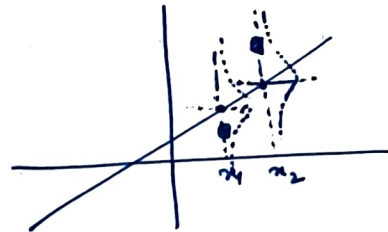
R.T.F :- Likelihood or log-likelihood when $e \sim \text{Bernoulli}(p)$

For a datapoint

$$P(e|w; x) = p^e (1-p)^{1-e} \quad \text{where } e \in \{0, 1\}$$

this is equivalent to

$$P(y|w; x) = p^{y-w^T x} (1-p)^{1-y+w^T x} \quad (0.5)$$



Conventional way

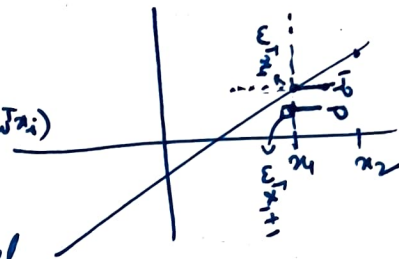
• $\rightarrow$ pred value $f(x; w)$ or $w^T x$

■ $\rightarrow$ act value y

For a bunch of sample $D = \{x_1, x_2, \dots, x_n\}$ or $\{x_i\}_{i=1}^N$ which are i.i.d and corresponding output is y_i

$$\text{Thus } p(\{y_i\}_{i=1}^N | w, \{x_i\}_{i=1}^N) = \prod_{i=1}^N p^{y_i - w^T x_i} (1-p)^{1-y_i + w^T x_i}$$

$$= p^{\sum_{i=1}^N (y_i - w^T x_i)} (1-p)^{N - \sum_{i=1}^N (y_i - w^T x_i)} \quad (0.5)$$



(0.5) { Is assuming $e \sim \text{Bernoulli}(p)$ a good model for std. regression? The answer is No, because y and $w^T x$, the actual and predicted values, are continous. Thus $e = y - w^T x$ must almost always be a continous value. However, modelling e as $\text{Bernoulli}(p)$ will make it discrete ($e \in \{0, 1\}$). This causes inconsistency!

$$\begin{aligned} \text{Thus } y &= e + f(x; w) \\ \text{or } y &= e + w^T x \\ \Rightarrow y &\in \{w^T x, w^T x + 1\} \text{ as } e \in \{0, 1\} \end{aligned}$$

Solution for w (Find $\arg\max_w \mathcal{L}(w)$)

$$\mathcal{L}(w) = \underset{\text{(likelihood)}}{P(\{y_i\}_{i=1}^N | w, \{x_i\}_{i=1}^N)} = \prod_{i=1}^N p^{y_i - w^T x_i} (1-p)^{1 - y_i + w^T x_i} = p^{\sum_{i=1}^N (y_i - w^T x_i)} (1-p)^{N - \sum_{i=1}^N (y_i - w^T x_i)}$$

$$\mathcal{L}(w) = \ln \left(p^{\sum_{i=1}^N (y_i - w^T x_i)} (1-p)^{N - \sum_{i=1}^N (y_i - w^T x_i)} \right)$$

$$\mathcal{L}(w) = \sum_{i=1}^N (y_i - w^T x_i) \ln p + \left(N - \sum_{i=1}^N (y_i - w^T x_i) \right) \ln(1-p)$$

$$\nabla_w \mathcal{L}(w) = \left(-\sum_{i=1}^N x_i \right) \ln p + \left(\sum_{i=1}^N x_i \right) \ln(1-p)$$

$$= \left(\sum_{i=1}^N x_i \right) \ln \left(\frac{1-p}{p} \right) = 0$$

As we see, w has vanished from the equation. Thus there is no solution

0.5

0.25

Q4)

Ans) Given:- 1) Original

1.1) Ridge Reg is an MAP estimate

1.2) Gaussian likelihood of error

1.3) Gaussian prior on weights

2) Modified, only prior on weights changes to MVB

R.T.F :- MAP obj function for w

We know that since we have gaussian likelihood on error for say N samples (all i.i.d), we get following likelihood

$$P(y|x;w) = \prod_{i=1}^N \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(y_i - w^T x_i)^2}{2\sigma^2}} \quad \text{where } y \rightarrow \text{actual value,}$$

$w^T x \rightarrow \text{predicted value}$

and $e \sim N(0, \sigma^2)$ or

$y \sim N(w^T x, \sigma^2)$

$$P(Y|X;w) = \prod_{i=1}^N \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2} \frac{(y_i - w^T x_i)^2}{\sigma^2}} \quad \text{--- (0.5)}$$

$$Y = \{y_i\}_{i=1}^N \quad \& \quad X = \{x_i\}_{i=1}^N$$

* This assumption is very crucial to be stated and explicitly ~~is~~ isn't stated in question. Deduct 0.25 if not stated!

$$P(Y|X; \omega) = (2\pi\sigma^2)^{-N/2} e^{-\frac{1}{2\sigma^2} \sum_{i=1}^N (y_i - \omega^T x_i)^2}$$

$P(\omega)$ is prior over parameters and is ~~assumed~~ sampled from multi-variate Bernoulli distribution (thus $\omega_j \sim \text{Bernoulli}(p_j)$ where $j \in \{1, 2, \dots, d+1\}$ when x is a d -dimensional datapoint).

$$P(\omega) = \prod_{j=1}^{d+1} p(\omega_j) = \prod_{j=1}^{d+1} p_j^{\omega_j} (1-p_j)^{1-\omega_j} \quad (\text{assuming all } \omega_j \text{'s are i.i.d.})$$

Thus $P(\omega|X; Y) = P(Y|X; \omega) \cdot P(\omega)$ (ignoring ω independent terms such as probability w.r.t x 's and y 's)

$$= ((2\pi\sigma^2)^{-N/2} e^{-\frac{1}{2\sigma^2} \sum_{i=1}^N (y_i - \omega^T x_i)^2 / \sigma^2}) \prod_{j=1}^{d+1} p_j^{\omega_j} (1-p_j)^{1-\omega_j}$$

If we assume that $p_j = p$ for all $j \in \{1, 2, \dots, d+1\}$

$$P(\omega|X; Y) = ((2\pi\sigma^2)^{-N/2} e^{-\frac{1}{2\sigma^2} \sum_{i=1}^N (y_i - \omega^T x_i)^2 / \sigma^2}) p^{\sum \omega_j} (1-p)^{d+1 - \sum \omega_j}$$

$$= \underbrace{((2\pi\sigma^2)^{-N/2} e^{-\frac{1}{2\sigma^2} \sum_{i=1}^N (y_i - \omega^T x_i)^2 / \sigma^2})}_{\text{const. ignore it!}} \left(\frac{p}{1-p} \right)^{\sum \omega_j} \quad (\text{term } (1-p)^{d+1} \text{ is a const w.r.t } \omega. \text{ Hence ignored}).$$

$$\ln P(\omega|X; Y) = \underbrace{-\frac{N}{2} \ln 2\pi\sigma^2}_{\text{const. ignore it!}} - \frac{1}{2\sigma^2} \sum_{i=1}^N (y_i - \omega^T x_i)^2 + \left(\sum_{j=1}^{d+1} \omega_j \right) \ln \left(\frac{p}{1-p} \right)$$

Hence, the obj function is $\max_{\omega} P(\omega|X; Y)$ or $\max_{\omega} \ln P(\omega|X; Y)$.

Q5)

Ans) Derive the bias-variance decomposition given that

i) $y = f(x) + \epsilon$

ii) $\epsilon \sim U(-a, a)$

iii) Expected squared prediction error = $E[(y - \hat{f}(x))^2]$

$$e = E[(y - \hat{f}(x))^2] \quad (\text{where expectation is w.r.t } \epsilon \text{ and samples})$$

$$e = E[(f(x) + \epsilon - \hat{f}(x))^2]$$

$$= E[(f(x) - \hat{f}(x))^2] + E[\epsilon^2] - 2E[(f(x) - \hat{f}(x))\epsilon]$$

$$= E[(f(x) - \hat{f}(x))^2] + E[\epsilon^2] - 2E[(f(x) - \hat{f}(x))]E[\epsilon]$$

(as $f(x) - \hat{f}(x)$ and ϵ are independent)

$$E[\epsilon] = 0 \quad (E[\epsilon] \text{ is expectation of noise}) \text{ and}$$

$$E[\epsilon^2] \text{ is the irreducible noise!}$$

$$e = E[(f(x) - \hat{f}(x))^2] + E[\epsilon^2]$$

We have taken expectation w.r.t ϵ . Now we must expectation w.r.t samples (D).

$$\bar{f} = E_D[\hat{f}(x)]$$

$$e = E[(f(x) - \bar{f}(x) + \bar{f}(x) - \hat{f}(x))^2] + E[\epsilon^2]$$

$$= E[(f(x) - \bar{f}(x))^2] + E[(\bar{f}(x) - \hat{f}(x))^2] + 2E[(f(x) - \bar{f}(x))(\bar{f}(x) - \hat{f}(x))]$$

Since $f(x)$ is the true generating function & $\bar{f}(x)$ is my trained model. Hence, its difference must be bias

$$E[(f(x) - \bar{f}(x))^2] \rightarrow \text{bias}^2$$

Since $\bar{f}(x)$ is average of $\hat{f}(x)$ w.r.t D . Thus $E[(\hat{f}(x) - \bar{f}(x))^2]$, is variance

We know that $E[\epsilon^2]$ (the irreducible error) = $\int_{-\infty}^{\infty} p(\epsilon) \epsilon^2 d\epsilon$

$$\text{As } \epsilon \sim U(-a, a), E[\epsilon^2] = \int_{-a}^a p(\epsilon) \epsilon^2 d\epsilon = \int_{-a}^a \frac{\epsilon^2}{2a} d\epsilon = \frac{1}{2a} \frac{\epsilon^3}{3} \Big|_{-a}^a$$

$$= \frac{1}{6a} (a^3 - (-a)^3)$$

$$= \frac{a^3}{3a} = \frac{a^2}{3}$$

Q6)

Ans) Degree 2 has high bias. Hence, it is an underfit model. — (0.25)

Degree 5 has low bias and low variance. This is the optimal model. — (0.25)

Degree 9 has very low bias but very high variance. Hence, it is an overfit model. — (0.25)

Hence, Degree 5 is the best model from bias-variance tradeoff point of view. — (0.25)

Q7)

Ans) a) In classical region, a very complex model overfits (low bias and high variance).

In overparameterization too, the model generalizes (has great train-test accuracy) but it is not always guaranteed. Hence, False.
crucial point! (0.5)

b) ^{const} Regularization averts overfitting as it ^{adds} constraints on model complexity. Hence, regularization generally reduces variance is a "True" statement ——— (0.5)